

Code-Layout, Readability and Reusability

Date	24 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of structure, readability, maintainability, and reusability. The project follows a modular architecture with separate frontend, backend, AI integration, and database layers

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing maintained	Yes	Standard indentation followed
2	Proper File Structure	Logical folder organization	Yes	Frontend, backend, AI and DB separated
3	Meaningful Variable Names	Clear naming conventions	Yes	Self-explanatory names
4	Function / Method Names	Descriptive functions	Yes	Coding standards followed
5	Code Comments	Important logic documented	Partial	Comments added for key modules
6	Modular Design	Reusable modules	Yes	Independent components
7	No Redundant Code	Duplicate code removed	Yes	Reusable utilities used
8	Error Handling	Validation & exceptions	Yes	Robust API handling

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User Authentication Module	FastAPI, Python	Login & Registration	High
2	AI Settlement Recommendation Module	Python, Gemini API	Debt Analysis	High

3	Negotiation Letter Generator	Python, Gemini API	Letter Generation	High
4	Financial Health Analysis Module	Python	Dashboard	High
5	Loan Management Module	FastAPI, SQLAlchemy	Loan Management	High
6	Database Module	SQLite, SQLAlchemy ORM	Data Storage	High
7	REST API Services	FastAPI	Frontend Communication	High
8	Dashboard UI Components	React.js, HTML, CSS, JavaScript	Frontend	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized architecture
Readability	5	Easy to understand and maintain
Reusability	5	Reusable modules implemented
Documentation / Comments	4	Adequate comments provided
Overall Score	5	High-quality and maintainable codebase